

Literature Review

COVID-19 Vaccination and Mortality: An Analytical Examination

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Introduction

The spread of COVID-19 started early before an emergency pandemic was called, right around 2019. On a global scale, it was a full pandemic during March 2020. This was the biggest pandemic we've seen in the modern era. The medical facilities had a major crisis on having enough materials and space to take care of all the patients on life support from respiratory problems - one of the many critical symptoms from COVID-19. Because of this, there was a race to come out with a COVID-19 vaccine that would slow down the spread of the virus. Getting a standard vaccine approved by the FDA takes 5-10+ years, but since this case scenario was an emergency pandemic, the process was sped up.

My secondary research findings hit three major points: the global impact of Covid-19 vaccination campaigns on mortality outcomes, the waning effectiveness of vaccine protection over time, and the social-cognitive factors that shaped individual vaccination behavior. These equations are directly tied to the secondary data analysis using Our World in Data COVID-19 datasets and the primary survey research through Qualtrics given to adults 18 years old or older. These two datasets are the foundation of my analytical project.

Data Quality Limitations

A major challenge in analyzing COVID-19 mortality data are the inconsistencies with how deaths are reported in various countries. Ezech identified several structural barriers to accurate COVID-19 death reporting in low- and middle-income countries, including wide variations in testing policies, limited laboratory capacity, and reliance on hospital-based death reporting that systematically excludes community deaths (Feyissa et al., 2021). Countries differ on what

definitions they chose to adopt, whether it be WHO-aligned definitions, leaving international death statistics inaccurate and limited.

Keeping all these inconsistencies in mind concerning the reporting deaths in various countries, keeping the analysis focused on countries with a civil registration and accurate death reporting system was a priority. The United States, United Kingdom, India, Mexico and China, but I did keep global data for the visualization. This rule of thumb is consistent in the secondary data (Our World in Data, 2024).

Vaccination and Global Mortality Reduction

The best comprehensive data on the impact of the COVID-19 death rate is from (Watson et al., 2022). He made a mathematical transmission model and applied it to all 185 countries. The analysis showed that since the global vaccine deployment averted about 19.8 million deaths, within the first year (Dec 2020-Dec 2021). He also found without the vaccination deployment; the modeled death count would have been 31.4 million during that first year. An important thing to not is he found most of the averted deaths from the vaccine where in lower-middle-income countries, where lower medical quality is prevalent. This also shows making the vaccine free and available on a global level was beneficial.

All these findings provide great context for the secondary data being analyzed. Both the vaccination and COVID-19 deaths datasets, and both from Our World in Data (2024), showing vaccination counts through the pandemic along with reported death counts. Putting these together, allows for exploratory analysis of the relationships derived from vaccination rollout time and observed death trends. This project is not as complex and timely as (Watson et al.'s 2022)

epidemiological model, but his findings serve as a benchmark against the secondary/primary data I've gathered to be observed and interpreted against.

Vaccine Effectiveness

Another important opponent to touch on is the vaccines effectiveness over a course of time. Feikin created a systematic review and meta regression of studies looking at vaccine effectiveness (Feikin et al. 2022). They looked at various COVID-19 variants and vaccine types and found vaccine protection was high against death in the initial months post-vaccination. Within six months of post-vaccination, the symptomatic illness declined by 20-30 percent. This could be due to pathogens and strands changing overtime. Mentioning it declined the symptoms of the illness instead of severe outcomes like death is important. This distinction is used when interpreting mortality data because a declining case prevention does not imply a declining death protection.

These findings both correlate with my secondary data and my survey used for collecting primary data. My survey has questions that touch on vaccination doses received (Q7) and the odds of the person getting future booster shots (Q8). This allows me to examine whether or not the patients that got more doses of the vaccine, meaning consistent protection against COVID-19 (much like what Feikin's findings proved) had different attitudes than the people that got less doses of the vaccine. The secondary data also touches on whether vaccination momentum correlates with death rate trends across counties over time.

Behavioral Determinants

Beyond clinical outcomes, understanding why people have chosen to decline the vaccination is key to completing the analysis concerning of the pandemic response. A longitudinal study traced 2,663 people to examine their social and psychological predictors of vaccine behavior...will they take the vaccine or will they not. The study found the decision to get the vaccine was correlated to someone's political beliefs. People that leaned more conservative, conspiracy beliefs, and low levels of trust were the strongest negative predictors ($p < .05$). People that were more democratic, higher levels of trust in the government, more thoughts about possible health risks, and social conformity had high positive predictors of vaccination trends (Han et al. 2023). This proves vaccination behavior is not predicted based off health calculations, it's embedded in social standings.

The survey used for collecting primary data is for collecting behavioral dimensions. Question 12 asks about personal health concerns, government, news media, social media, and advice from medical professionals, using a seven-point scale, strongly disagree strongly agree. This instrument defines what social constructs are identified in (Han et al., 2023) in my primary data collection, making a direct comparison between my survey respondents influences and the broader behavior patterns from the documented literature, using more responses.

Question 16 in the survey is special, it's an open-ended question that invites any respondent to describe in detail of their vaccination decision and what influenced it, this is more qualitative data. This allows more than just fixed responses and allows for a richer response to add to qualitative findings. This primary data validates any existing behavioral research on vaccine behaviors and adds original data to understanding local vaccination attitudes around Eastern Illinois University.

Excess Mortality

It's important to have a baseline as well. This was used to identify "excess mortality" deaths beyond the baseline level of a period. This is a bit more reliable way of seeing the impact the pandemic has caused. The secondary dataset from Our World Data (2024) is updated often, so an up to date reading of excess mortality deaths and COVID-19 deaths against all mortality trends. Doing this cross-validation in metrics shows the deaths that are not directly from the pandemic. Even deaths that were not recorded as COVID-19 deaths.

This project has a baseline or reference point, and this is the excess mortality rate. Any country that reports deaths that stray from the baseline are immediately suspicious of bad quality in data that Feyissa et al (2021) described in further detail. These workarounds that have to be done just prove how difficult it is to work with real-world public health data. Measures we have to take to see which data is reliable to base analysis on.

Research Questions

Three major questions asked for this project:

1. Did COVID-19 vaccination rollout correlate with reductions in mortality?
2. Which countries were hit hardest, and when did vaccines arrive?
3. What does the global and continental picture look like over time?

Appendix

I used Microsoft SQL server to analyze the secondary data and used Tableau for my visualizations. My dashboard in Tableau answers the first two research question, vaccine rollout, death percentage, number of cases, and infection rates overtime. My primary data collection

through the survey done in Qualtrics given to people 18 or older addresses question 3. Looking at qualitative data, discovering social attitudes towards vaccination behavior stemming from political beliefs or personal medical risks.

Data cleaning steps taken in SQL server start with:
Removing Continent level rows: Changed continent name to country name in rows. They now have a NULL value in the column; this prevents them from being double counted as deaths. Using:
WHERE continent IS NOT NULL

Fixing Types:

The total_deaths and new_deaths columns were NVARCHAR, not numeric types.

Using: CAST(total_deaths AS BIGINT)

Fixing cases with zero:

A lot of countries early on the pandemic had zero cases. Dividing total_deaths by total_cases caused an error and crash.

Using: CAST(total_deaths AS FLOAT) / NULLIF(total_cases, 0) * 100

New Vaccination totals:

Instead of using the daily report of vaccination every day. I used a rolling cumulative total instead using SUM() to order it by date.

Using: SUM(CONVERT(INT, vac.new_vaccinations)) OVER (PARTITION BY dea.location ORDER BY dea.data) AS RollingPeopleVaccinated

Focusing in on five countries:

Many countries had reports for vaccination records, inconsistent records. Getting accurate data with a good civil registration system I used only five countries discussed earlier in the review.

Primary	Data	Cleanup:
Removed 7 previews, 4 incompletes, 1 under-18, and the 2 blanks. This came out to be 40 clean responses.		

Conclusion

Looking at the literature that's been out I can confidently say the release of the COVID-19 vaccine has had a positive impact on reducing mortality at a global scale. I was hoping the relationship on the vaccine and protection was simple, but its complex and time sensitive. This project looked at both publicly available secondary data and original primary data to see the relationships of the pandemic on a global and individual scale. The queries made through SQL show classic application of business analytics and points out public health challenges when it comes to analyzing raw or clean data.

What I learned

- Raw data requires alot of cleaning and scraping. Alot of extra data are in datasets that you don't need it's just extra. Keeping the data as simplistic as possible helps organize it and view it.
- Making SQL queries is a lot of trial and error. Leave this process a lot of time especially since I am newer to SQL. It's good practice, but it also takes time. Also keep things organized as you do it, keeping headers and labels for each section you work on help remember what the SQL was supposed to show.

- Primary data can be more effective the sooner you start it. Having time for respondents and getting as much data as possible makes analyzing it more fun and informal. Making sure you have zero grammar errors in the survey is important, read it 10 times. I published it and had a spelling error and quickly fixed it but was embarrassed on who could have seen it, and I was hoping it didn't make them think the survey was unprofessional.
- Data just standing alone is not enough; you have to have frameworks. Framing research question well helps show the correlation between my vaccination rollouts and mortality trends. Also being honest about what issues you ran into is important to because there are natural limits. It's better to address it than act like it's nonexistent.
- You can never do enough reading on past literature for a literature review. At some point you must pick up the most convincing information that correlates with your data because you could be reading forever and get lost in it. Staying on track is important as well; it's easy to get sidetracked and answer questions that you didn't initially have set up to answer.

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